

# Un poco de estadística con Python en Rstudio

```
#!pip install pandas
```

```
# import libraries for loading dataset
import pandas as pd
import numpy as np
# import libraries for plotting
import matplotlib.pyplot as plt
import seaborn as sns
from matplotlib import rcParams
#%matplotlib inline
plt.style.use('dark_background')
# ignore warnings
import warnings
warnings.filterwarnings('ignore')
```

```
# loading the data set
df = pd.read_csv("adult.csv", encoding='latin-1')
print(df.head())
```

|   | age | workclass | fnlwgt | education    | education.num | marital.status | \ |
|---|-----|-----------|--------|--------------|---------------|----------------|---|
| 0 | 90  | ?         | 77053  | HS-grad      | 9             | Widowed        |   |
| 1 | 82  | Private   | 132870 | HS-grad      | 9             | Widowed        |   |
| 2 | 66  | ?         | 186061 | Some-college | 10            | Widowed        |   |
| 3 | 54  | Private   | 140359 | 7th-8th      | 4             | Divorced       |   |
| 4 | 41  | Private   | 264663 | Some-college | 10            | Separated      |   |

|   | occupation        | relationship  | race  | sex    | capital.gain | \ |
|---|-------------------|---------------|-------|--------|--------------|---|
| 0 | ?                 | Not-in-family | White | Female | 0            |   |
| 1 | Exec-managerial   | Not-in-family | White | Female | 0            |   |
| 2 | ?                 | Unmarried     | Black | Female | 0            |   |
| 3 | Machine-op-inspct | Unmarried     | White | Female | 0            |   |

|   |                |           |       |        |   |
|---|----------------|-----------|-------|--------|---|
| 4 | Prof-specialty | Own-child | White | Female | 0 |
|---|----------------|-----------|-------|--------|---|

|   | capital.loss | hours.per.week | native.country | income |
|---|--------------|----------------|----------------|--------|
| 0 | 4356         | 40             | United-States  | <=50K  |
| 1 | 4356         | 18             | United-States  | <=50K  |
| 2 | 4356         | 40             | United-States  | <=50K  |
| 3 | 3900         | 40             | United-States  | <=50K  |
| 4 | 3900         | 40             | United-States  | <=50K  |

```
df.shape
```

```
(32561, 15)
```

```
df.columns
```

```
Index(['age', 'workclass', 'fnlwgt', 'education', 'education.num',
      'marital.status', 'occupation', 'relationship', 'race', 'sex',
      'capital.gain', 'capital.loss', 'hours.per.week', 'native.country',
      'income'],
      dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 32561 entries, 0 to 32560
Data columns (total 15 columns):
#   Column                Non-Null Count  Dtype
---  -
0   age                   32561 non-null  int64
1   workclass             32561 non-null  object
2   fnlwgt                32561 non-null  int64
3   education             32561 non-null  object
4   education.num         32561 non-null  int64
5   marital.status        32561 non-null  object
6   occupation            32561 non-null  object
7   relationship          32561 non-null  object
8   race                  32561 non-null  object
9   sex                   32561 non-null  object
10  capital.gain          32561 non-null  int64
11  capital.loss          32561 non-null  int64
```

```

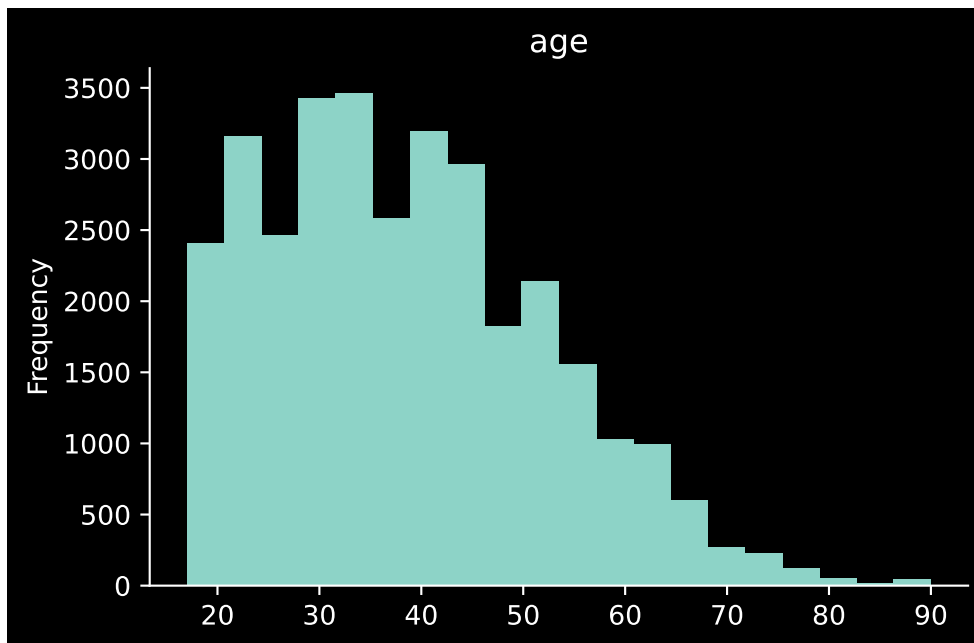
12  hours.per.week  32561 non-null  int64
13  native.country  32561 non-null  object
14  income          32561 non-null  object
dtypes: int64(6), object(9)
memory usage: 3.7+ MB

```

```

from matplotlib import pyplot as plt
df['age'].plot(kind='hist', bins=20, title='age')
plt.gca().spines[['top', 'right']].set_visible(False)
plt.show()

```



```
print(df.tail())
```

|       | age | workclass | fnlwgt | education    | education.num | marital.status     | \ |
|-------|-----|-----------|--------|--------------|---------------|--------------------|---|
| 32556 | 22  | Private   | 310152 | Some-college | 10            | Never-married      |   |
| 32557 | 27  | Private   | 257302 | Assoc-acdm   | 12            | Married-civ-spouse |   |
| 32558 | 40  | Private   | 154374 | HS-grad      | 9             | Married-civ-spouse |   |
| 32559 | 58  | Private   | 151910 | HS-grad      | 9             | Widowed            |   |
| 32560 | 22  | Private   | 201490 | HS-grad      | 9             | Never-married      |   |

|       | occupation      | relationship  | race  | sex    | capital.gain | \ |
|-------|-----------------|---------------|-------|--------|--------------|---|
| 32556 | Protective-serv | Not-in-family | White | Male   | 0            |   |
| 32557 | Tech-support    | Wife          | White | Female | 0            |   |

|       |                   |           |       |        |   |
|-------|-------------------|-----------|-------|--------|---|
| 32558 | Machine-op-inspct | Husband   | White | Male   | 0 |
| 32559 | Adm-clerical      | Unmarried | White | Female | 0 |
| 32560 | Adm-clerical      | Own-child | White | Male   | 0 |

|       | capital.loss | hours.per.week | native.country | income |
|-------|--------------|----------------|----------------|--------|
| 32556 | 0            | 40             | United-States  | <=50K  |
| 32557 | 0            | 38             | United-States  | <=50K  |
| 32558 | 0            | 40             | United-States  | >50K   |
| 32559 | 0            | 40             | United-States  | <=50K  |
| 32560 | 0            | 20             | United-States  | <=50K  |

```
# As we observe education and education_num both are same , so we can drop one of the column
df.drop(['education.num'], axis = 1, inplace = True)
df.drop(['race'], axis = 1, inplace = True)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 32561 entries, 0 to 32560
Data columns (total 13 columns):
#   Column                Non-Null Count  Dtype
---  -
0   age                    32561 non-null  int64
1   workclass              32561 non-null  object
2   fnlwgt                 32561 non-null  int64
3   education              32561 non-null  object
4   marital.status         32561 non-null  object
5   occupation             32561 non-null  object
6   relationship           32561 non-null  object
7   sex                    32561 non-null  object
8   capital.gain           32561 non-null  int64
9   capital.loss           32561 non-null  int64
10  hours.per.week         32561 non-null  int64
11  native.country         32561 non-null  object
12  income                 32561 non-null  object
dtypes: int64(5), object(8)
memory usage: 3.2+ MB
```

```
print(df.describe().T)
```

| count | mean | std | min | 25% | \ |
|-------|------|-----|-----|-----|---|
|-------|------|-----|-----|-----|---|

|                |         |               |               |         |          |
|----------------|---------|---------------|---------------|---------|----------|
| age            | 32561.0 | 38.581647     | 13.640433     | 17.0    | 28.0     |
| fnlwgt         | 32561.0 | 189778.366512 | 105549.977697 | 12285.0 | 117827.0 |
| capital.gain   | 32561.0 | 1077.648844   | 7385.292085   | 0.0     | 0.0      |
| capital.loss   | 32561.0 | 87.303830     | 402.960219    | 0.0     | 0.0      |
| hours.per.week | 32561.0 | 40.437456     | 12.347429     | 1.0     | 40.0     |

|                | 50%      | 75%      | max       |
|----------------|----------|----------|-----------|
| age            | 37.0     | 48.0     | 90.0      |
| fnlwgt         | 178356.0 | 237051.0 | 1484705.0 |
| capital.gain   | 0.0      | 0.0      | 99999.0   |
| capital.loss   | 0.0      | 0.0      | 4356.0    |
| hours.per.week | 40.0     | 45.0     | 99.0      |

```
df.nunique()
```

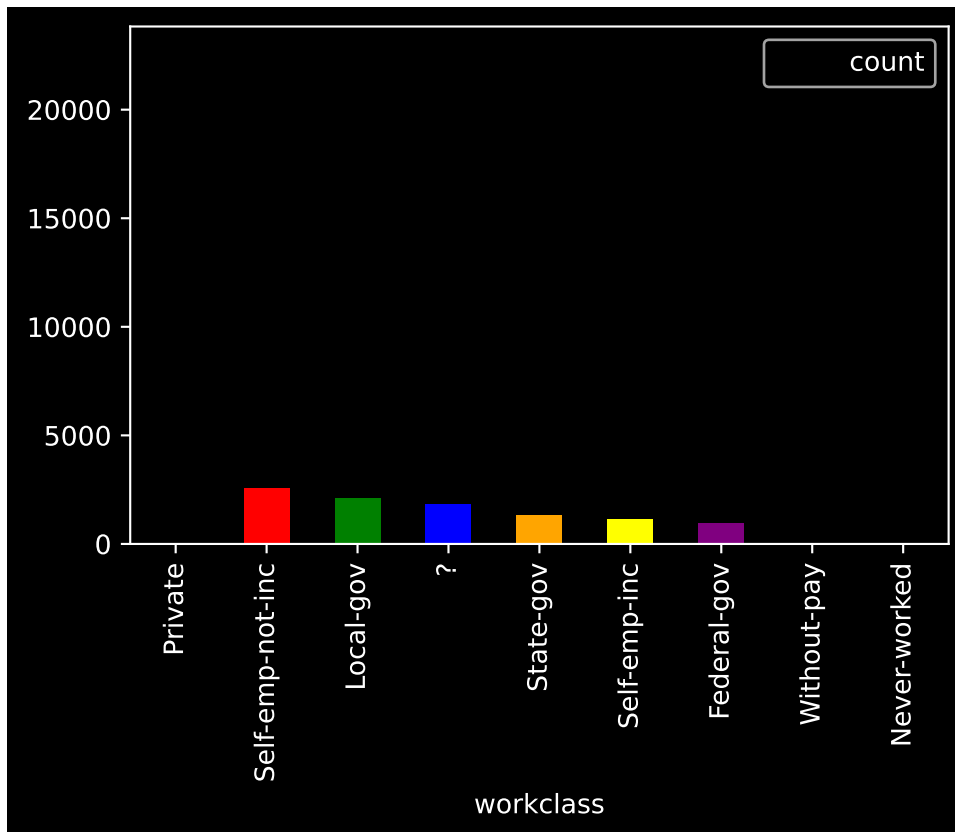
```
age          73
workclass     9
fnlwgt    21648
education    16
marital.status  7
occupation   15
relationship   6
sex           2
capital.gain  119
capital.loss   92
hours.per.week  94
native.country  42
income        2
dtype: int64
```

```
df['age'].value_counts()
```

```
age
36    898
31    888
34    886
23    877
35    876
...
83     6
88     3
```

```
85      3
86      1
87      1
Name: count, Length: 73, dtype: int64
```

```
df['workclass'].value_counts()
colors = ["black","red", "green", "blue", "orange", "yellow", "purple"]
df.workclass.value_counts().plot.bar(color=colors,legend=True)
plt.show()
```



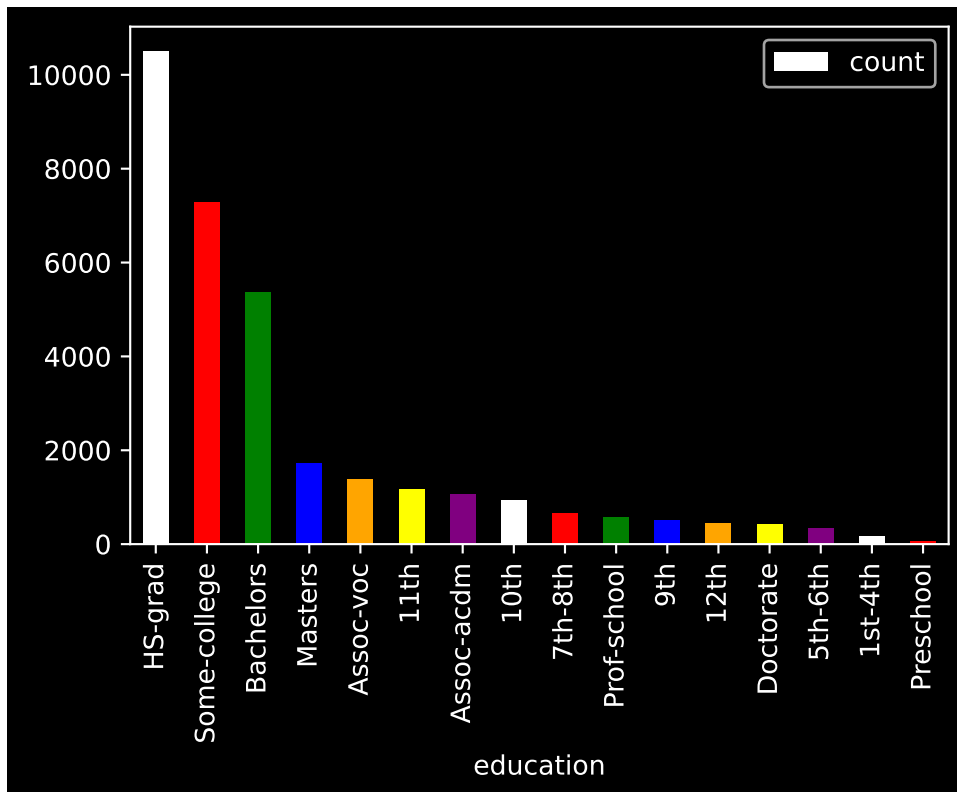
```
df['education'].value_counts()
```

```
education
HS-grad      10501
Some-college  7291
Bachelors    5355
Masters       1723
```

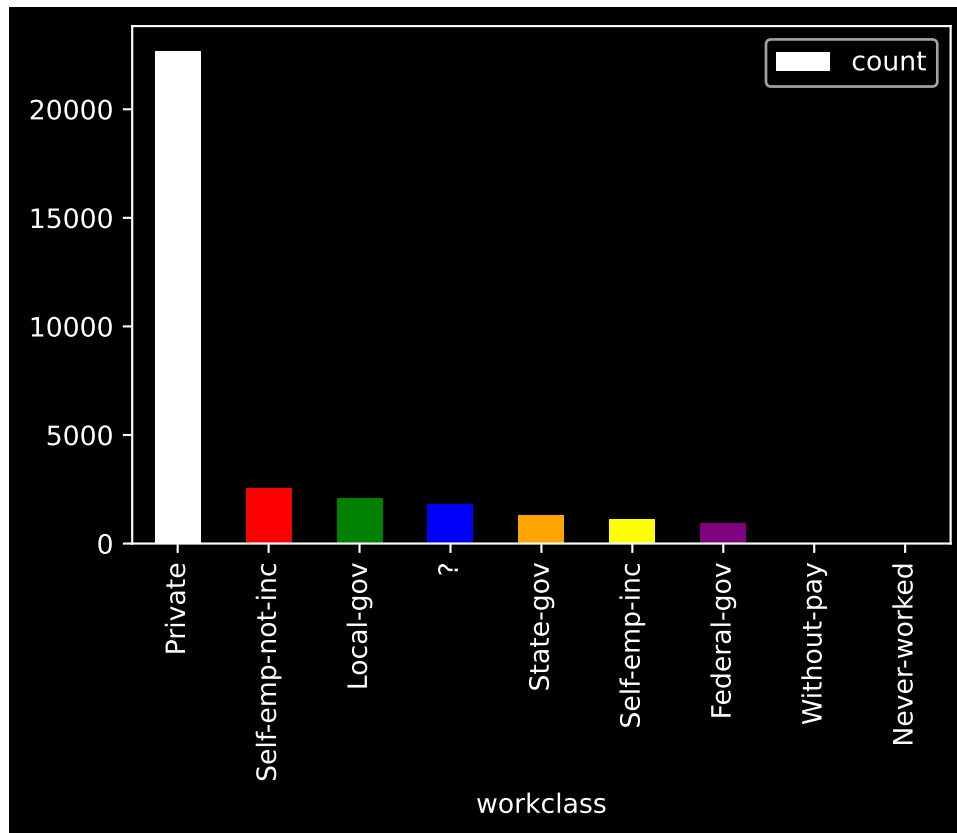
|             |      |
|-------------|------|
| Assoc-voc   | 1382 |
| 11th        | 1175 |
| Assoc-acdm  | 1067 |
| 10th        | 933  |
| 7th-8th     | 646  |
| Prof-school | 576  |
| 9th         | 514  |
| 12th        | 433  |
| Doctorate   | 413  |
| 5th-6th     | 333  |
| 1st-4th     | 168  |
| Preschool   | 51   |

Name: count, dtype: int64

```
df['education'].value_counts()
colors = ["white","red", "green", "blue", "orange", "yellow", "purple"]
df.education.value_counts().plot.bar(color=colors,legend=True)
plt.show()
```



```
df['workclass'].value_counts()
colors = ["white","red", "green", "blue", "orange", "yellow", "purple"]
df.workclass.value_counts().plot.bar(color=colors,legend=True)
plt.show()
```



```
df['hours.per.week'].value_counts()
```

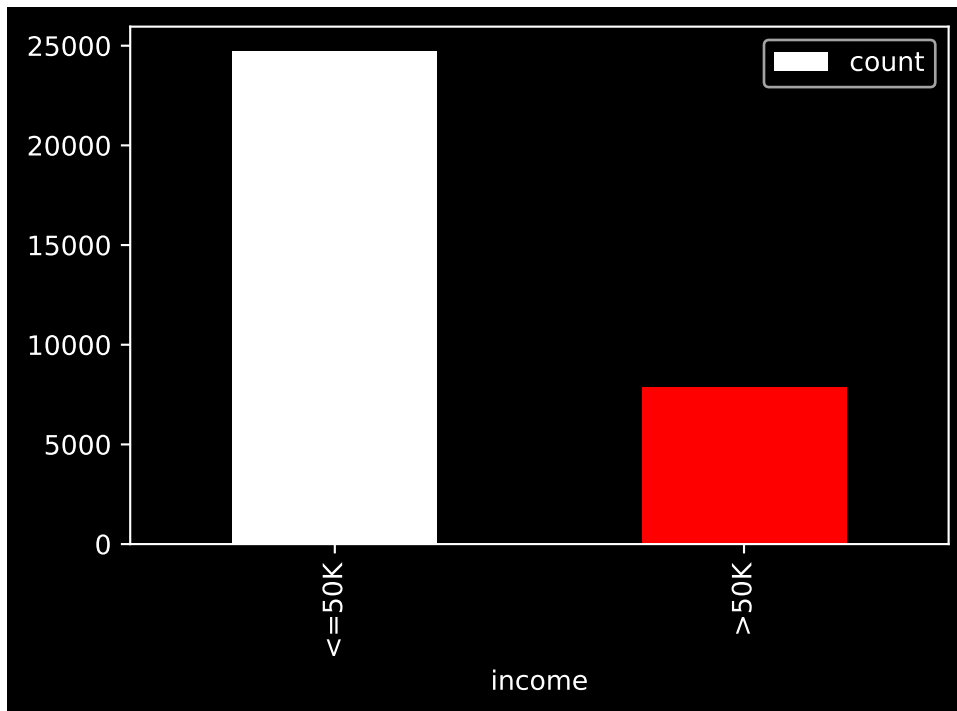
```
hours.per.week
40    15217
50     2819
45     1824
60     1475
35     1297
...
94         1
82         1
92         1
```

```
87      1
74      1
Name: count, Length: 94, dtype: int64
```

```
df['income'].value_counts()
```

```
income
<=50K    24720
>50K      7841
Name: count, dtype: int64
```

```
df['income'].value_counts()
colors = ["white", "red", "green", "blue", "orange", "yellow", "purple"]
df.income.value_counts().plot.bar(color=colors, legend=True)
plt.show()
```



```
df.isna().sum()
```

```
age      0
workclass 0
```

```

fnlwgt      0
education   0
marital.status  0
occupation   0
relationship 0
sex          0
capital.gain 0
capital.loss 0
hours.per.week 0
native.country 0
income      0
dtype: int64

```

```
df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 32561 entries, 0 to 32560
Data columns (total 13 columns):
#   Column          Non-Null Count  Dtype
---  -
0   age             32561 non-null  int64
1   workclass       32561 non-null  object
2   fnlwgt          32561 non-null  int64
3   education       32561 non-null  object
4   marital.status  32561 non-null  object
5   occupation      32561 non-null  object
6   relationship     32561 non-null  object
7   sex             32561 non-null  object
8   capital.gain    32561 non-null  int64
9   capital.loss    32561 non-null  int64
10  hours.per.week  32561 non-null  int64
11  native.country  32561 non-null  object
12  income          32561 non-null  object
dtypes: int64(5), object(8)
memory usage: 3.2+ MB

```

```

#catogrical column
catogrical_column = [column for column in df.columns if df[column].dtypes=='object']
print(catogrical_column)

```

```
['workclass', 'education', 'marital.status', 'occupation', 'relationship', 'sex', 'native.country']
```

```
#numerical_column
numerical_column = [column for column in df.columns if df[column].dtypes != 'object']
print(numerical_column)
```

```
['age', 'fnlwgt', 'capital.gain', 'capital.loss', 'hours.per.week']
```

```
print(df.head())
```

|   | age | workclass | fnlwgt | education    | marital.status | occupation        | \ |
|---|-----|-----------|--------|--------------|----------------|-------------------|---|
| 0 | 90  | ?         | 77053  | HS-grad      | Widowed        | ?                 |   |
| 1 | 82  | Private   | 132870 | HS-grad      | Widowed        | Exec-managerial   |   |
| 2 | 66  | ?         | 186061 | Some-college | Widowed        | ?                 |   |
| 3 | 54  | Private   | 140359 | 7th-8th      | Divorced       | Machine-op-inspct |   |
| 4 | 41  | Private   | 264663 | Some-college | Separated      | Prof-specialty    |   |

|   | relationship  | sex    | capital.gain | capital.loss | hours.per.week | \ |
|---|---------------|--------|--------------|--------------|----------------|---|
| 0 | Not-in-family | Female | 0            | 4356         | 40             |   |
| 1 | Not-in-family | Female | 0            | 4356         | 18             |   |
| 2 | Unmarried     | Female | 0            | 4356         | 40             |   |
| 3 | Unmarried     | Female | 0            | 3900         | 40             |   |
| 4 | Own-child     | Female | 0            | 3900         | 40             |   |

|   | native.country | income |
|---|----------------|--------|
| 0 | United-States  | <=50K  |
| 1 | United-States  | <=50K  |
| 2 | United-States  | <=50K  |
| 3 | United-States  | <=50K  |
| 4 | United-States  | <=50K  |

```
#univariate analysis
plt.figure(figsize=(10,10))
sns.histplot(data=df['education'],kde=True,legend=True)
plt.show()
```

